

Lecture 5

□ Electric Field Due to a Charged Disk

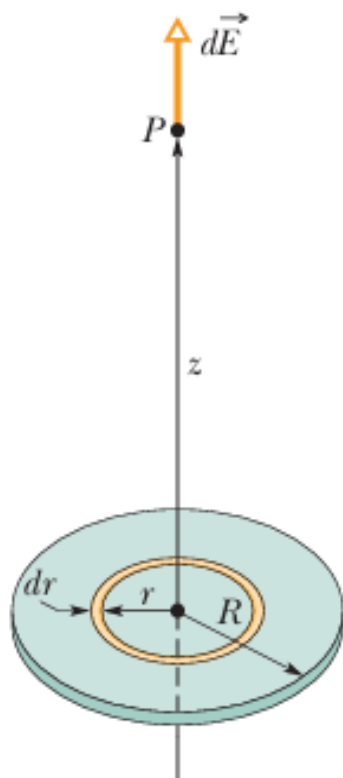


Figure: A disk of radius R and uniform positive charge. The ring shown has radius r and radial width dr . It sets up a differential electric field $d\vec{E}$ at point P on its central axis.

$$dE = \frac{dq z}{4\pi\epsilon_0(z^2 + r^2)^{3/2}}.$$

$$E = \int dE = \frac{\sigma z}{4\epsilon_0} \int_0^R (z^2 + r^2)^{-3/2} (2r) dr,$$

$$E = \frac{\sigma z}{4\epsilon_0} \left[\frac{(z^2 + r^2)^{-1/2}}{-\frac{1}{2}} \right]_0^R.$$

$$E = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{z}{\sqrt{z^2 + R^2}} \right) \quad (\text{charged disk})$$

$$dq = \sigma dA = \sigma (2\pi r dr).$$

$$\int X^m dX = \frac{X^{m+1}}{m+1},$$

If we let $R \rightarrow \infty$ while keeping z finite, the second term in the parentheses in Eq. 22-26 approaches zero, and this equation reduces to

$$E = \frac{\sigma}{2\epsilon_0} \quad (\text{infinite sheet}). \quad (22-27)$$

This is the electric field produced by an infinite sheet of uniform charge located on one side of a nonconductor such as plastic. The electric field lines for such a situation are shown in Fig. 22-4.

We also get Eq. 22-27 if we let $z \rightarrow 0$ in Eq. 22-26 while keeping R finite. This shows that at points very close to the disk, the electric field set up by the disk is the same as if the disk were infinite in extent.

□ A point charge in an Electric Field

Here we begin to determine what happens to a charged particle when it is in an electric field set up by other stationary or slowly moving charges. What happens is that an electrostatic force acts on the particle, as given by

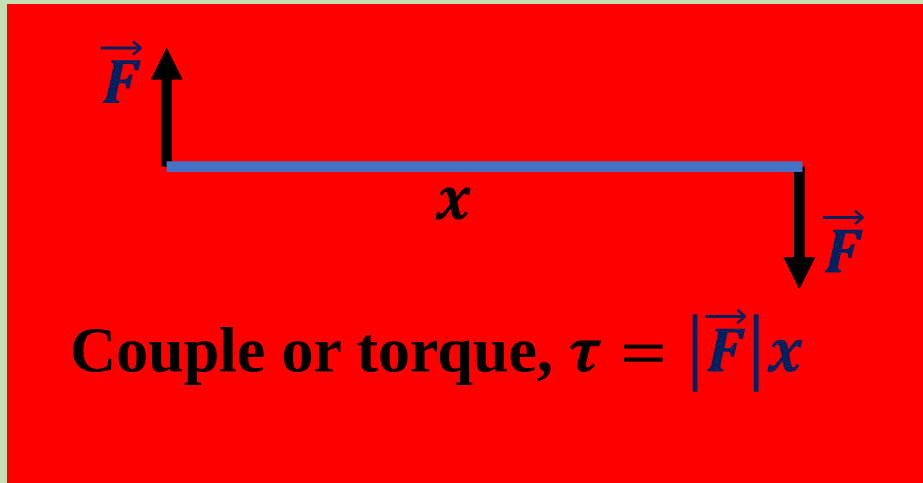
$$\vec{F} = q\vec{E}, \quad (22-28)$$

in which q is the charge of the particle (including its sign) and $\vec{E}$ is the electric field that other charges have produced at the location of the particle. (The field is *not* the field set up by the particle itself; to distinguish the two fields, the field acting on the particle in Eq. 22-28 is often called the *external field*. A charged particle or object is not affected by its own electric field.) Equation 22-28 tells us



The electrostatic force $\vec{F}$ acting on a charged particle located in an external electric field $\vec{E}$ has the direction of $\vec{E}$ if the charge q of the particle is positive and has the opposite direction if q is negative.

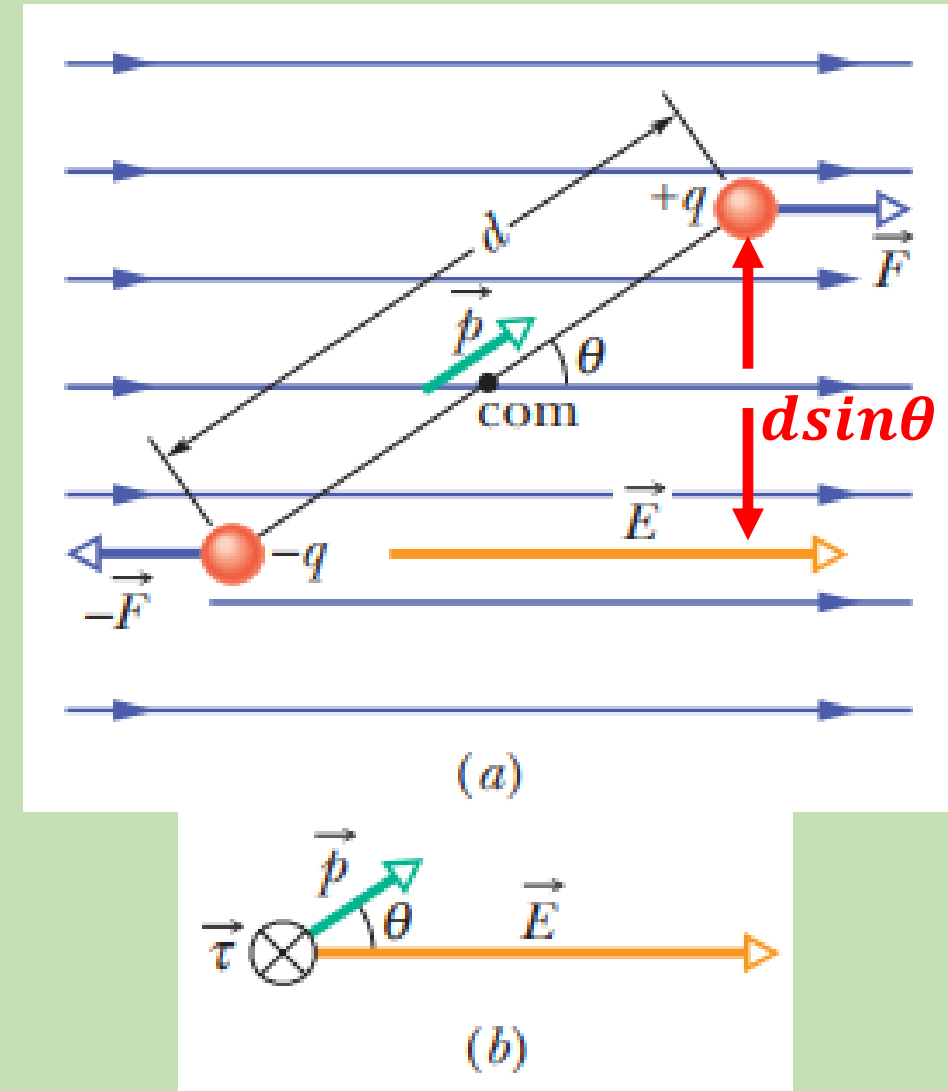
□ A Dipole in an Electric Field



From the right side figure (a),
The torque , $\tau = |\vec{F}|x$; where, $x = d\sin\theta$

$$\begin{aligned}\text{Therefore, } \tau &= |\vec{F}|d\sin\theta \\ &= qEd\sin\theta \\ &= (qd)E\sin\theta \\ &= pE\sin\theta\end{aligned}$$

$$\vec{\tau} = \vec{p} \times \vec{E}$$



❖ Potential Energy of an Electric Dipole

Let us consider the work done by the external electric field in turning the dipole from an initial angle θ_0 to a final angle θ is

$$w = \int dw = \int_{\theta_0}^{\theta} \vec{\tau} \cdot \overrightarrow{d\theta} = \int_{\theta_0}^{\theta} -\tau d\theta$$

Where, $\vec{\tau}$ is the torque exerted by the external electric field. The minus sign is necessary because the torque, $\vec{\tau}$ tends to decrease the angle, θ .

$$w = \int_{\theta_0}^{\theta} -pE \sin\theta d\theta = -pE \int_{\theta_0}^{\theta} \sin\theta d\theta = pE(\cos\theta - \cos\theta_0).$$

Since the electrostatic force is conservative force, so
work done by the agent + work done by the external electric field = 0

$$U + w = 0$$

$$\text{Or, } U = -w$$

$$U = -pE(\cos\theta - \cos\theta_0).$$

We arbitrarily define the reference angle to be $\theta_0 = 90^\circ$, then at any angle θ the potential energy is

$$U = -pE\cos\theta.$$

$$U = -\vec{p} \cdot \vec{E}$$

Thus, U is a minimum when $\vec{p}$ and $\vec{E}$ is parallel.

Thank You